The historical CH density instrument

Control: Match of Historical to Modern Geographic Units — Historical CH density index (State level).

1. Describe historical density within a modern state. This State-level instrument uses historical population and historical reporting-area geometry. Let P_{jt} be the population of reporting area j in year t, and a_{jt} its mapped area in square kilometres. A reporting area is usually one historical county; counties with a joint population report can form one combined area. Each area is assigned to the modern state containing more than half of its area. The assignment must be unique.

2. Average log densities using population shares. Let $\mathcal{J}_{g,t}$ contain the historical reporting areas assigned to modern state g within the selected historical scope. The instrument is

$$z_{g,t} = \sum_{j \in \mathcal{J}_{g,t}} \frac{P_{jt}}{\sum_{k \in \mathcal{J}_{g,t}} P_{kt}} \log\left(\frac{P_{jt}}{a_{jt}}\right).$$

A reporting area with twice the population receives twice the weight. The calculation uses whole reporting-area populations and densities, rather than allocating each population by the fraction of area inside the state. Zero-population areas contribute zero weight. The total assigned population must be positive.

3. This is separate from the modern CH index. The historical index is constructed before estimating the modern regression. No estimated modern elasticity enters its construction. The regression uses the displayed weighted mean of log densities directly. The Levels–Natural log selector does not transform it again. A negative value can be valid when historical density is below one person per square kilometre.

The modern CH model, in contrast, combines current county employment and land area and estimates $\rho = \theta - 1$. The historical instrument supplies variation intended to identify that modern density elasticity; it is not itself the estimated elasticity.

4. Coverage and identification remain necessary. Eligible states need known populations, unique majority assignments and at least 95% coverage by the assigned historical areas. Known source omissions can make a state unavailable. The mapped historical area is polygon area, not a separate land-only measure. A reporting area may cross a modern state boundary even with a unique majority assignment. Finally, a usable instrument must predict the relevant modern density variation and satisfy the assumption that historical settlement affects modern productivity only through the modeled channel after controls. The construction alone does not establish either requirement.